ZARA Sales Data analysis EDA

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df= pd.read_csv("Business_sales_EDA.csv")
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20252 entries, 0 to 20251
Data columns (total 15 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Product ID        20252 non-null  int64
 1   Product Position  20252 non-null  object
 2   Promotion         20252 non-null  object
 3   Product Category  20252 non-null  object
 4   Seasonal          20252 non-null  object
 5   Sales Volume      20252 non-null  int64
 6   brand             20252 non-null  object
 7   name              20251 non-null  object
 8    price            20252 non-null  object
 9   currency          20252 non-null  object
 10  terms             20252 non-null  object
 11  section           20252 non-null  object
 12  season            20252 non-null  object
 13  material          20252 non-null  object
 14  origin            20252 non-null  object
dtypes: int64(2), object(13)
memory usage: 2.3+ MB
```

```
df.head()
```

```
   Product ID Product Position Promotion Product Category Seasonal  \
0      110075   Front of Store        No         clothing      Yes
1      110295          End-cap       Yes         clothing       No
2      110329   Front of Store       Yes         clothing      Yes
3      110805   Front of Store       Yes         clothing      Yes
4      111521   Front of Store       Yes         clothing      Yes

   Sales Volume brand
name  \
0           864  Zara                WOOL BLEND FELT TEXTURE JACKET

1          1312  Zara                           HOODED DENIM JACKET
```

```
2          1181  Zara                 CONTRAST SOLE LEATHER SNEAKERS

3          1255  Zara  FAUX LEATHER OVERSIZED JACKET LIMITED EDITION

4          1275  Zara                        WASHED TECHNICAL JACKET


     price  currency    terms section  season    material      origin
price
0   $19.99       USD  jackets   WOMAN  Autumn  Wool Blend    Portugal
19.99
1   $69.99       USD  jackets   WOMAN  Autumn      Cotton       India
69.99
2   $50.99       USD    shoes     MAN  Autumn   Polyester    Portugal
50.99
3   $21.99       USD  jackets     MAN  Autumn     Viscose       China
21.99
4   $16.95       USD  jackets     MAN  Autumn      Cotton  Bangladesh
16.95
```

```
df.describe()
```

```
          Product ID  Sales Volume         price
count   20252.000000  20252.000000  20252.000000
mean   208931.432303   1097.400454     41.949061
std      8961.076507    298.234609     23.380960
min    110075.000000    518.000000     12.000000
25%    204442.750000    849.000000     23.950000
50%    209505.500000    990.000000     35.950000
75%    214568.250000   1364.250000     53.950000
max    219631.000000   1940.000000    134.990000
```

```
df.isnull().sum().sort_values(ascending = False)
```

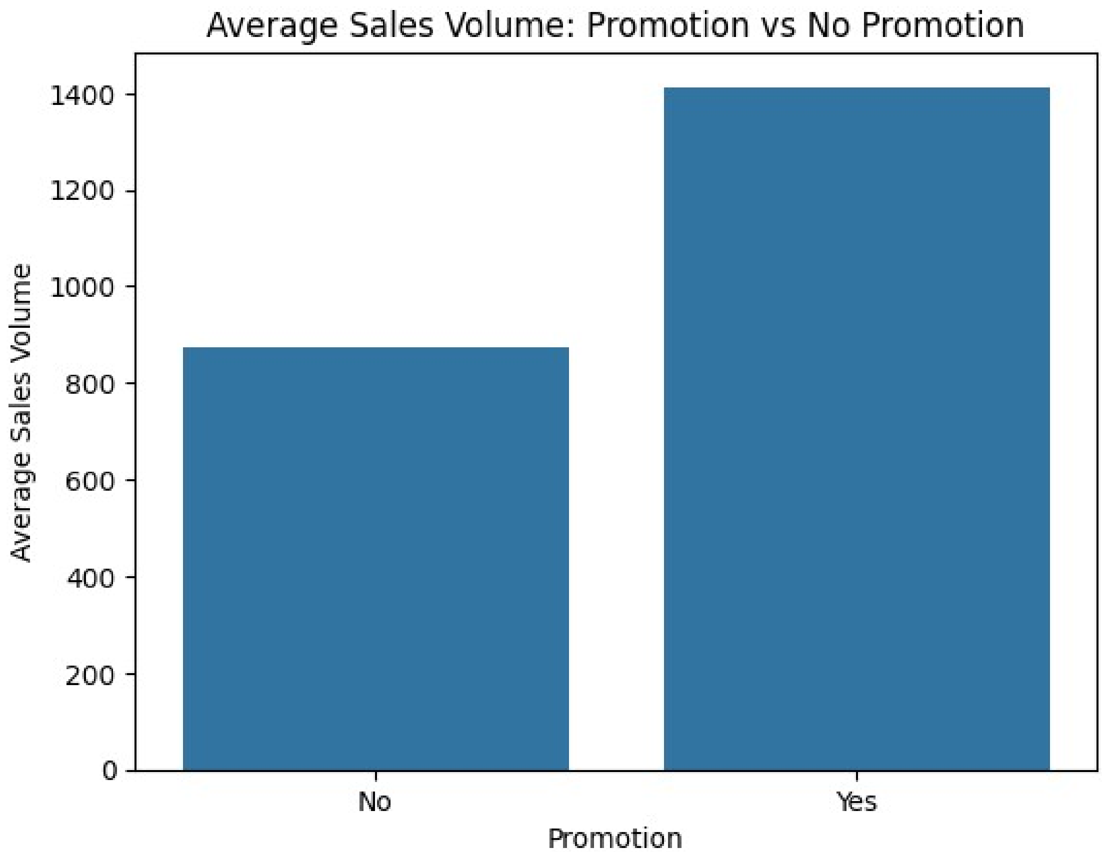
```
name                1
Product ID          0
Promotion           0
Product Position    0
Product Category    0
Seasonal            0
Sales Volume        0
brand               0
 price              0
currency            0
terms               0
section             0
season              0
material            0
origin              0
price               0
dtype: int64
```

```
df.columns

Index(['Product ID', 'Product Position', 'Promotion', 'Product
Category',
       'Seasonal', 'Sales Volume', 'brand', 'name', ' price ',
'currency',
       'terms', 'section', 'season', 'material', 'origin'],
      dtype='object')
```

Question-1;

Is there a noticeable difference in sales between promoted and non-promoted products?

```
promo_sales = df.groupby("Promotion")["Sales Volume"].mean()

plt.figure()
sns.barplot(x=promo_sales.index, y=promo_sales.values)
plt.title("Average Sales Volume: Promotion vs No Promotion")
plt.ylabel("Average Sales Volume")
plt.show()
```



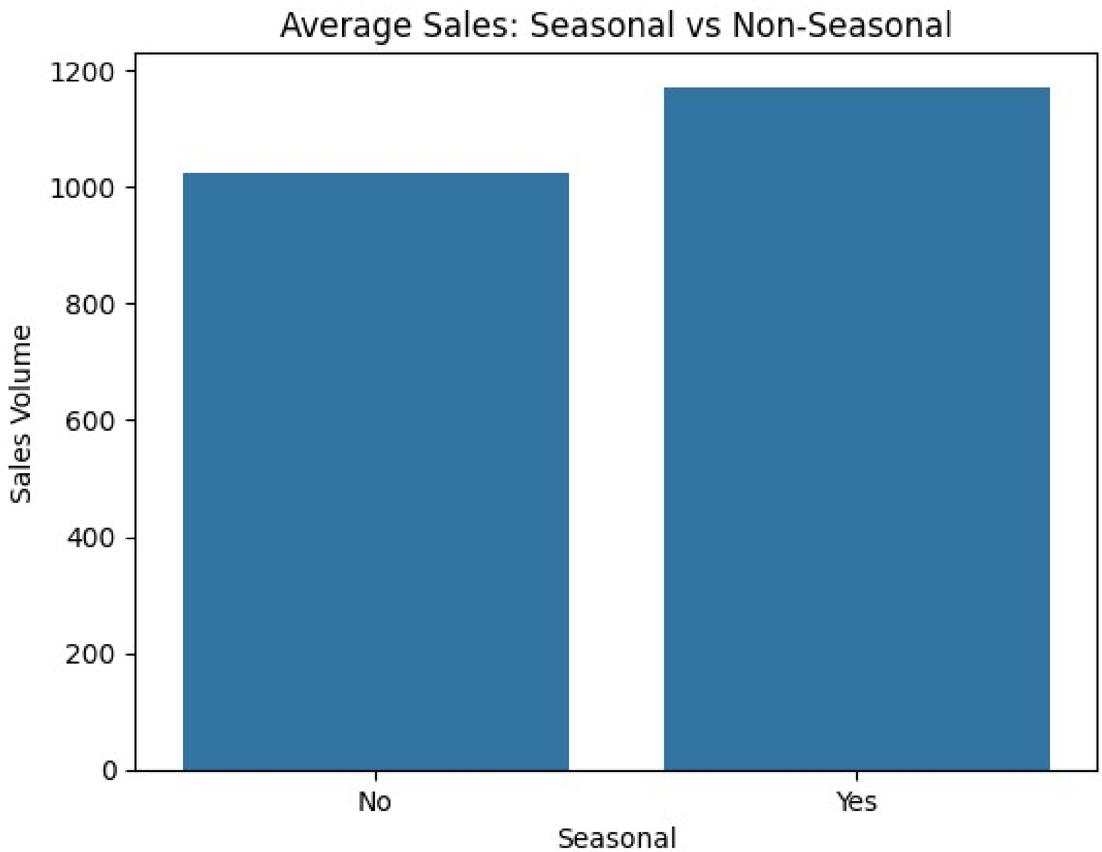
Answer;

Products under promotion show significantly higher average sales, confirming promotions are an effective sales strategy.

Question-2;

Does seasonality impact sales volume ;

```
seasonal_sales = df.groupby("Seasonal")["Sales Volume"].mean()

plt.figure()
sns.barplot(x=seasonal_sales.index, y=seasonal_sales.values)
plt.title("Average Sales: Seasonal vs Non-Seasonal")
plt.ylabel("Sales Volume")
plt.show()
```



Seasonal products outperform non-seasonal products, indicating higher consumer interest during seasonal trends.

Question -3;

Do higher-priced products sell less than lower-priced ones?

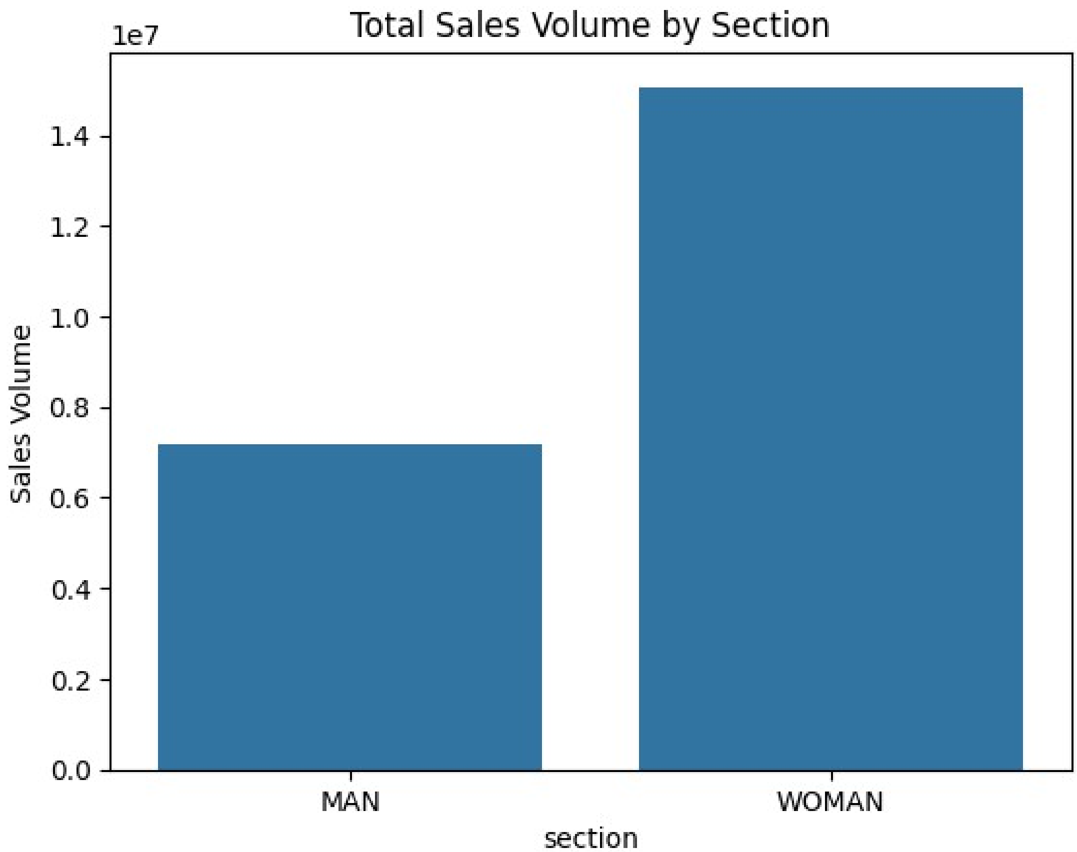
```
df["price"] = df[" price "].str.replace("$", "").astype(float)

plt.figure()
sns.scatterplot(x=df["price"], y=df["Sales Volume"])
plt.title("Price vs Sales Volume")
plt.xlabel("Price")
plt.ylabel("Sales Volume")
plt.show()
```

In here affordable price products sell more than high price products.

Question-4;

Which customer segment generates higher sales volume?

```
section_sales = df.groupby("section")["Sales Volume"].sum()

plt.figure()
sns.barplot(x=section_sales.index, y=section_sales.values)
plt.title("Total Sales Volume by Section")
plt.ylabel("Sales Volume")
plt.show()
```



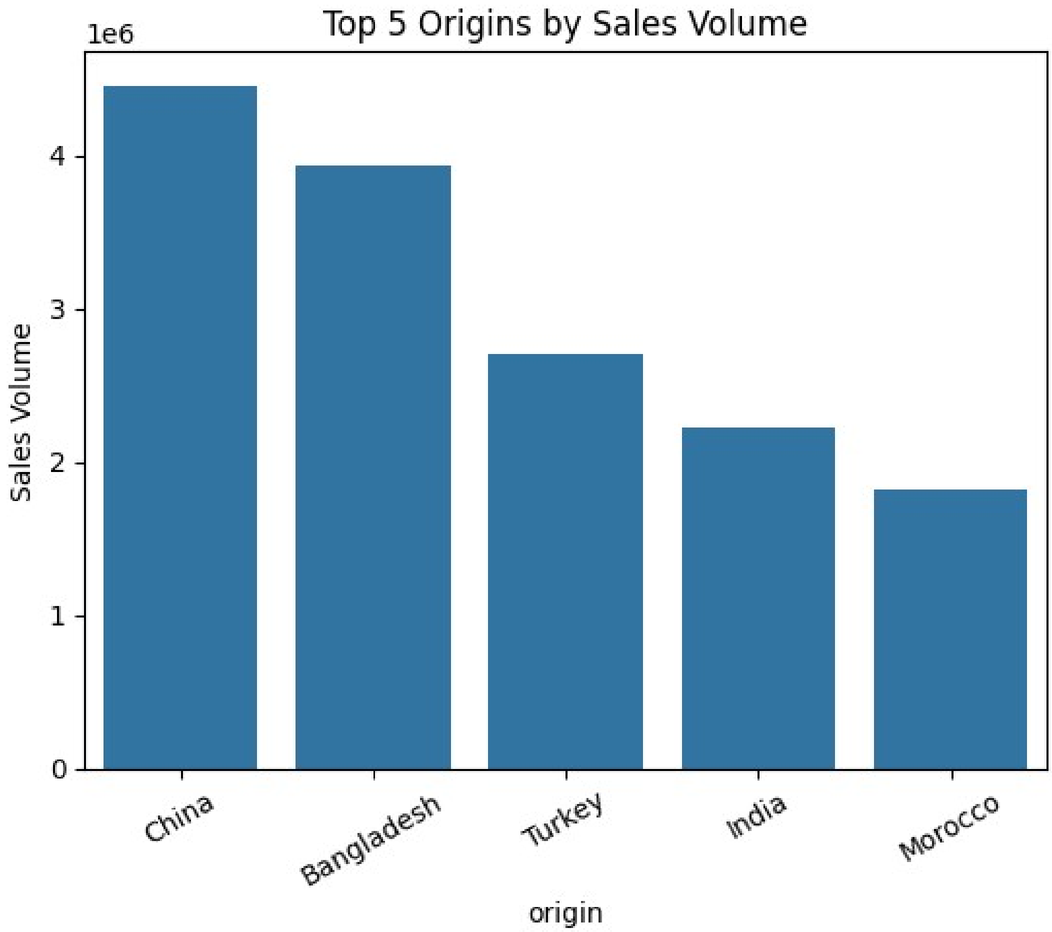
The WOMEN section records higher total sales, suggesting stronger purchase frequency and product variety

Question-5;

Does product origin influence sales performance?

```
origin_sales = df.groupby("origin")["Sales
Volume"].sum().sort_values(ascending=False).head(5)

plt.figure()
sns.barplot(x=origin_sales.index, y=origin_sales.values)
plt.title("Top 5 Origins by Sales Volume")
```

```
plt.ylabel("Sales Volume")
plt.xticks(rotation=30)
plt.show()
```



Products manufactured in China and Bangladesh contribute the highest sales volume, reflecting cost-effective production and high demand.

Conducted comprehensive Exploratory Data Analysis (EDA) on retail sales data using Pandas, NumPy, Matplotlib, and Seaborn.

Promotions and seasonal trends significantly increase sales performance, confirming Zara’s reliance on fast-fashion demand cycles.

Lower-priced products drive higher sales volume, emphasizing price sensitivity among customers.

The women’s category acts as the primary revenue engine, suggesting strategic prioritization.

Supply chain sourcing from cost-efficient countries supports competitive pricing and strong sales outcomes.

Business Value:

This analysis helps retail decision-makers optimize pricing strategy, promotional planning, inventory allocation, and supply chain sourcing to maximize revenue and demand responsiveness.